MOTOR INSURANCE MANAGEMENT SYSTEM

Introduction

The Motor Insurance Management System (MOTORAMATE) is a database-driven application developed to automate the complete motor insurance process. The system manages customer information, vehicle details, insurance quotations, premium calculations, payments, policy generation, and policy-related documents. It minimizes manual work, improves accuracy, and provides a secure and efficient way to manage motor insurance policies.

The system supports different types of users such as customers, brokers, sales agents, and administrators. It also allows customers to select insurance coverage plans, add optional coverages, make payments, and receive their insurance policy instantly after successful payment.

Objectives

- To automate the motor insurance process.
- To maintain customer and vehicle information securely.
- To calculate insurance premiums accurately.
- To generate insurance policies automatically.
- To manage broker commissions and prepaid balances.
- To generate policy schedules, credit notes, and debit notes.
- To reduce paperwork and improve operational efficiency.

System Requirements

Software Requirements

- Operating System: Windows 10/11
- Database: MySQL
- MySQL Workbench
- Visual Studio Code (Optional)
- Git & GitHub
- Microsoft Word (Documentation)

Hardware Requirements

- Processor: Intel Core i3 or above

- RAM: 4 GB or higher
- Storage: 20 GB Free Space
- Internet Connection

Types of Users

1. Administrator

- Manages the complete system.
- Creates master data.
- Manages users and brokers.
- Updates insurance products.
- Generates reports.

2. Customer

- Registers personal details.
- Adds vehicle information.
- Requests insurance quotations.
- Selects insurance coverage.
- Makes premium payments.
- Downloads insurance policies.

3. Broker

- Creates policies for customers.
- Uses prepaid credit balance for payments.
- Receives commission for policies sold.
- Tracks customer policies.

4. Sales Agent

- Assists customers in policy creation.
- Creates quotations.
- Updates customer information.
- Helps complete the insurance process.

Functional Requirements

The system should:

- Provide secure login with encrypted passwords.
- Store customer details.
- Store vehicle details.
- Maintain master data.
- Generate insurance quotations.
- Calculate premium automatically.
- Allow customers to select add-on coverages.
- Calculate GST.
- Support multiple payment methods.
- Support foreign currency payments.
- Generate unique policy numbers.
- Generate policy schedules.
- Generate credit and debit notes.
- Store all transactions securely.

Modules

Module 1 – User Management

- User Registration
- Login
- User Authentication
- Role Management

Module 2 – Vehicle Management

- Vehicle Make
- Vehicle Model
- Vehicle Category
- Vehicle Body Type

- Vehicle Color

Module 3 – Customer Management

- Personal Information
- Address Details
- Contact Information
- National ID

Module 4 – Quote Management

- Create Quote
- Select Coverage
- Premium Calculation
- Add-on Cover Selection

Module 5 – Payment Management

- Payment Processing
- Foreign Currency Payment
- Broker Credit Deduction

Module 6 – Policy Management

- Policy Generation
- Policy Schedule Generation
- Credit Note
- Debit Note

Database Tables

Master Tables

- Vehicle Make
- Vehicle Model
- Vehicle Category
- Vehicle Body
- Vehicle Color
- States

- Cities
- Product Configuration
- Add-on Covers
- Premium Rate
- LOV (List of Values)

Transaction Tables

- Users
- Login Users
- Brokers
- Customer Vehicle
- Quotes
- Quote Add-ons
- Payments
- Policies
- Broker Transactions
- Commission
- Credit Note
- Debit Note
- Policy Schedule

Process Flow

User Login

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Customer Registration

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Vehicle Registration

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Select Insurance Plan

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Choose Add-on Covers

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Premium Calculation

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GST Calculation

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Payment Processing

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Policy Generation

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Credit/Debit Note Generation

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Policy Schedule Generation

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Logout

Conclusion

The Motor Insurance Management System provides a reliable and efficient solution for managing the complete motor insurance process. It simplifies customer registration, vehicle management, quotation generation, premium calculation, payment processing, and policy generation. The system ensures data accuracy, improves operational efficiency, reduces manual work, and provides better customer service. This project demonstrates the practical implementation of a relational database using MySQL and can be further enhanced with web and mobile technologies for real-world deployment.